

Week 6

Hypothesis Testing

Paul Laskowski and D. Alex Hughes

UC Berkeley, School of Information

Speaker notes

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Speaker notes

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6.1 Historical Development of Frequentist Statistics

Introduction

- Hypothesis, H , is a model for how the world might work
- In practice, evidence is rarely conclusive
- We want $Pr(H|D)$, or the probability of the event that our hypothesis is true

The Dilemma

Whether our hypothesis is true is something we can never know

- The world is not a perfectly controlled lab
- Evidence collected contains information but does not begin to identify a unique model out of all the possible models
- We do not know how to weight all the possibilities out there

Example 1

Is there enough information to answer this? - How did the coin get there? - The context is missing: - How do we choose between the different models? - How do we weight all the alternatives? - Even with more information, you can never know the context completely

Example 2

- What is the probability that Newton's theory of gravity is correct?
- **Problem:** Newton's second theory seemed to work up to the precision of 17th-century instruments
- Only later were instruments developed that were precise enough to show Newton's laws were incorrect

Example 2 (cont.)

- How could Newton decide how probable his model was compared to general relativity (not imagined yet)?
- If he could have imagined another theory, would we be equipped to compare two very different ideas?
- We could never write down the infinite number of models consistent with observations of the planet in order to assign each a probability
- It may not even make sense to assign a probability to Newton's gravity (eg. true state of world or not)

Example 3

You discover three specimens of a new species of squid that measure 3.2, 3.3, and 4.0 feet long

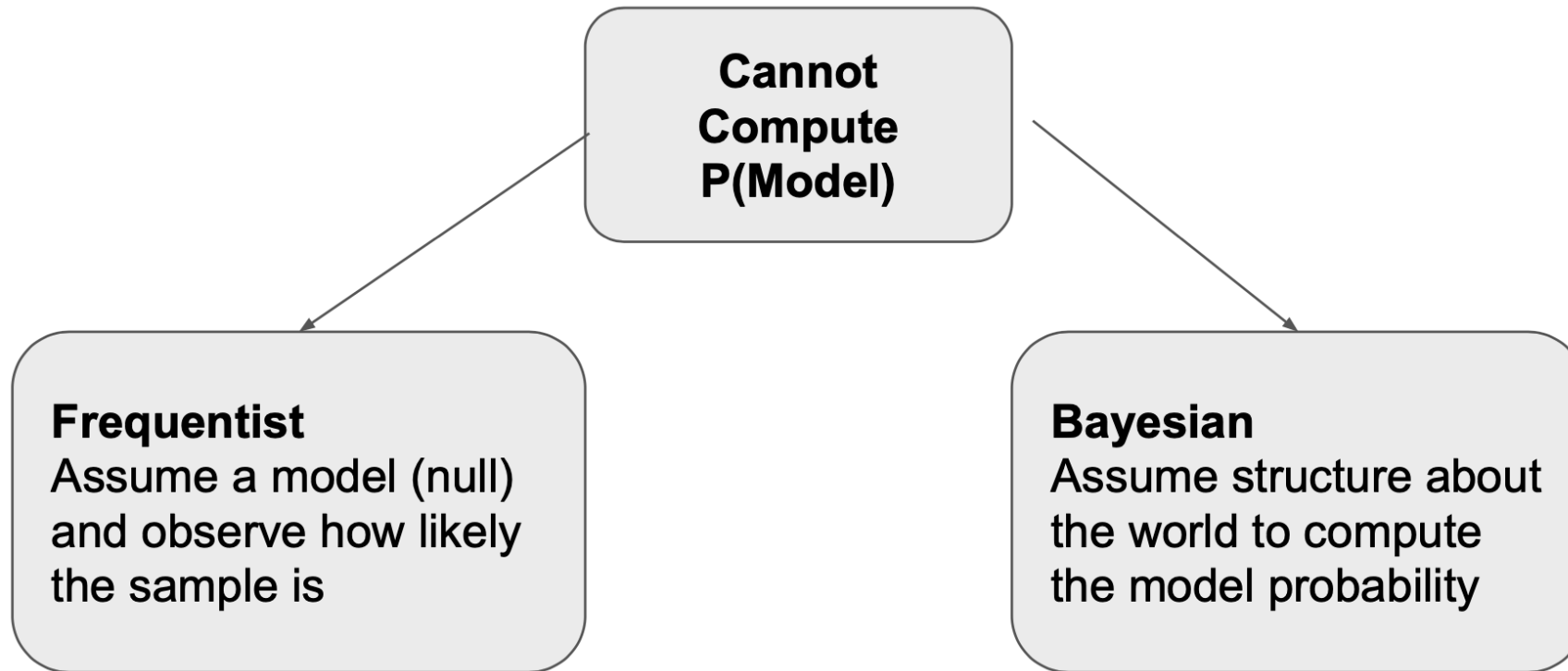
- What is the probability that the average length amount the entire species is 3.5 feet? - Probability zero for a single number (point estimate) probability that the average length is between 3 and 4 feet?
- Positive number for length
- No new numbers that have not been imagined
- A better grasp of the possibilities

Example 3 (cont.)

- However, we don't know how representative the specimens are - We still don't know all the relevant information
- Examples: deep water pressure, amount of light

We cannot deduce the probability of our model because we do not know enough about the structure of the world

Two Branches of Statistics



6.4 The Frequentist Approach

The Birth of Modern Statistics



Jerzy Neyman

April 15, 1894–



Egon Pearson

August 11, 1895–

The Central Dilemma

- We observe data, D
- Given that this data occurs, we want the probability that our hypothesis $P(H|D)$ is **true**

To a strict frequentist:

- Not just impossible to compute
- Does not even make sense to assign a probability to a hypothesis

Objective Probability

A frequentist defines probability as a matter of long-run frequencies

- We need to specify a collective of elements - Eg. throws of a dice
- **Collective:** a frame of observations that can happen over and over
- As number of observations approaches infinity, proportion of throws of the die that show a 3 is $1/6$
- The probability is the long-run frequency of the event relative to all observations (or of 3's relative to all throws) - Called **objective probability**

Objective Probability and Hypothesis

- If you view probability as objective, you cannot talk about the probability of a hypothesis
- it is just true or false
- **Subjective Probability:** The probability of a hypothesis - Allows for disagreement about what it is
- Reflects our lack of information

So, what probabilities can we study? - We need a long-run collective

$PR(D|H)$

- H = hypothesis
- D = data

Assume H is true and call it the null hypothesis

- Has to be quite specific (the only extra assumption we're making)
- Is the basis for predictions we need to make about what data should come out of the experiment
- Governs how the experiment behaves as we run it over and over

$PR(D|H)$ (cont.)

Now we have a meaningful collective

- Can look at the relative frequencies of different outcomes
- Can specifically look at the number of hypothetical experiments in which we would get data at least as extreme as D - Captured by p -value
- **-value:** The probability of getting data as extreme as our observations, assuming the null hypothesis is true

Components of a Hypothesis Test

Notes while Screensharing

[

$$g(x_1, X_2, \dots, X_n) \rightarrow S \in \mathbb{R}$$
$$H_0 = 0$$

]

Hypothesis Test Example

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- Let's use a stylized example to motivate the hypothesis testing framework.

Two Possible States of the World

Goal: Begin with a reasonable default supposition; leave this supposition behind if data provides compelling evidence

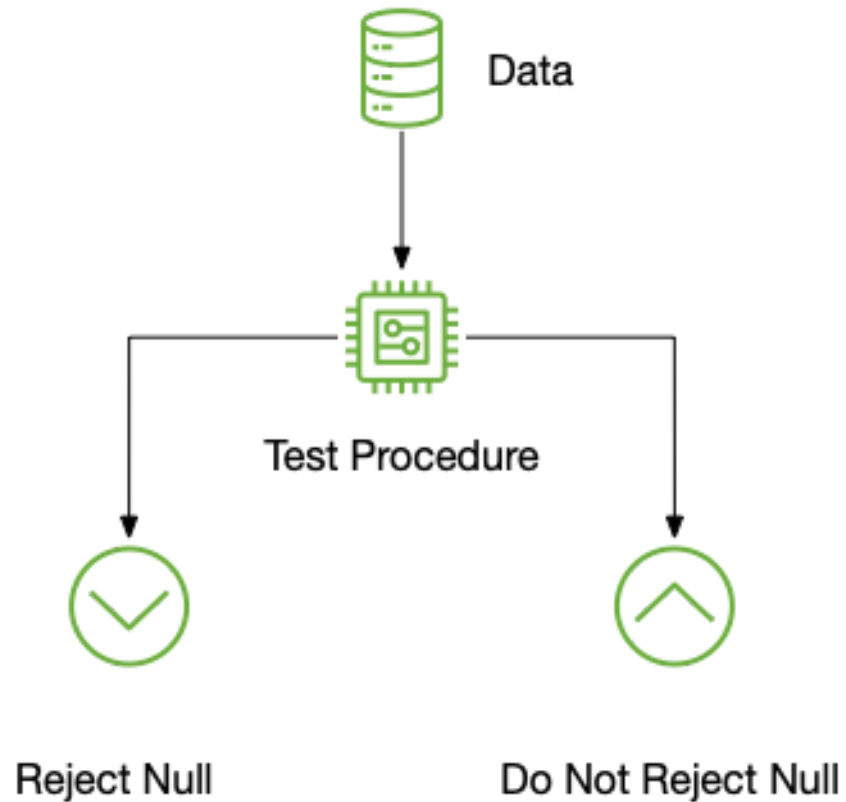
Null hypothesis- Default assumption, status quo, statement that data might overturn - H_0 : Usually $\mu = 0$ - No effect

Alternative hypothesis- Idea or alternative to status quo - H_a : Usually $\mu \neq 0$ - Some effect exists

With compelling evidence, we leave the specific null hypothesis (H_0) for the alternative (H_a)

A Hypothesis Test

A hypothesis test is a procedure.



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- This is very strict: you might want your hypothesis test to do more, to give you fine-grained information about the world. but statistics doesn't work like that. All we can do, is set up a specific null hypothesis and make this binary decision: reject or not reject.

False Positive and False Negative Errors

False Positive and False Negative Errors (cont.)

False Positive Errors - Typically the most destructive - Error rate, denoted α , is the probability of rejecting the null hypothesis when we should not; $P(\text{Reject } H_0 | H_0)$ - Starting with Ronald Fisher: set $\alpha = 0.05$

A hypothesis test is a procedure for rejecting or not rejecting a null, such that the false positive error rate is controlled ($\alpha = 0.05$).

Breaking Down a Test Procedure

A test statistic - A function of our sample - Measures deviations from the null hypothesis - Distribution must be completely determined by the null

A rejection region - A set of values for which we will reject the null - Chosen to be contrary to the null - Total probability must be $\alpha = 0.05$

What a Hypothesis Test Doesn't Do

A hypothesis test does not prove the null hypothesis. - We control Type 1 error rates - We cannot control Type 2 error rates - How can you be sure the real B is not 0.01? Or 0.00001?

Never accept the null hypothesis. - The valid decisions are reject and fail to reject.

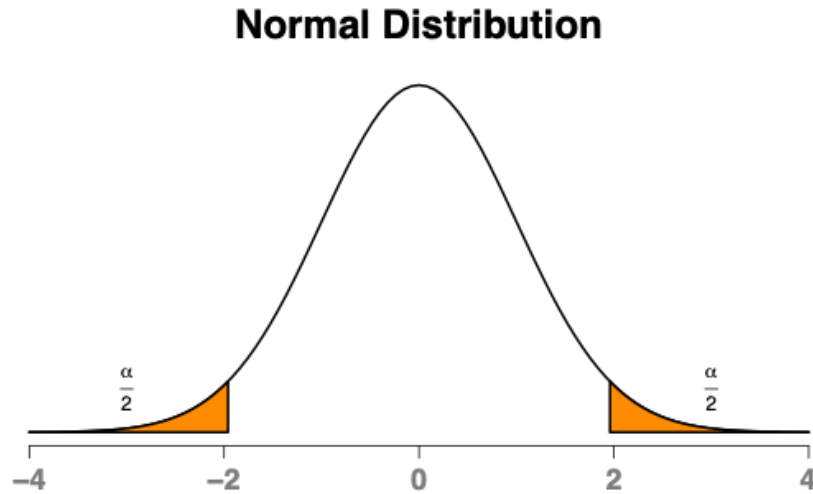
The One-Sample z-Test

Speaker notes

- We want to know if Vitamin W has an effect. so we write
- We need a statistic, and a rejection region
- For our statistic, we usually want to find something that follows a famous distribution.
- Can use the standardized mean.
- For our rejection region , we need to choose a subset of the real numbers. I'll show you the most common choice, above 1.96 and below -1.96.
- Why these numbers? If you integrate, you get a probability of 0.05.
- Ex:

One- and Two-Tailed Tests

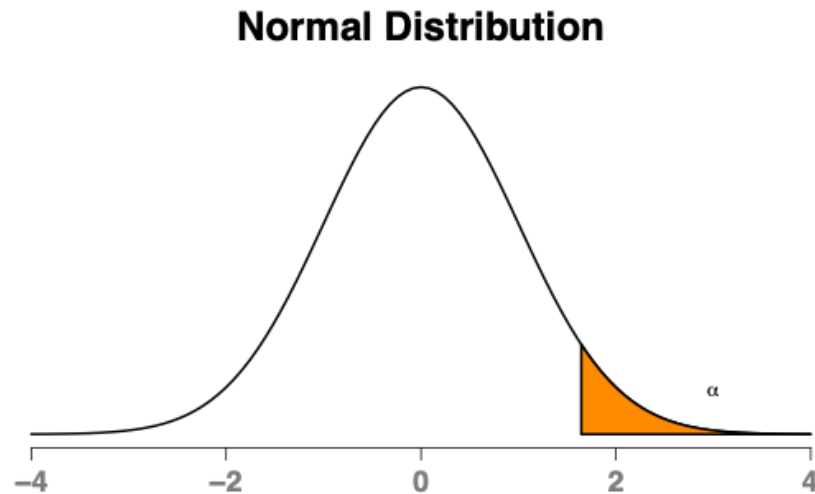
The Two-Tailed z-Test



$\mu = 0$ - Alternative hypothesis : $\mu \neq 0$

- Null hypothesis :

The One-Tailed z-Test

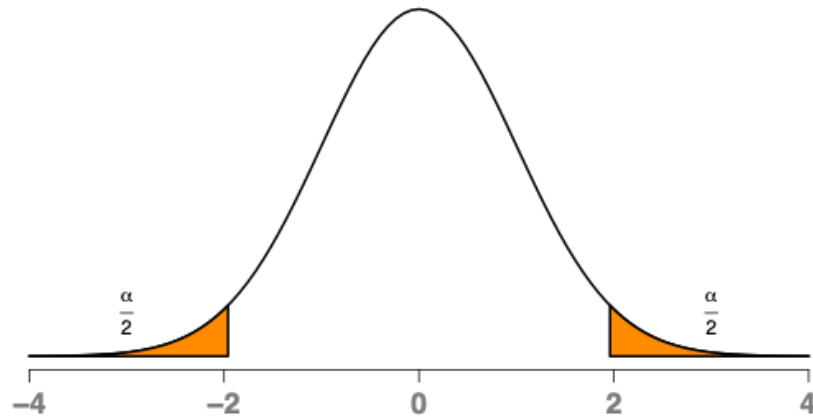


- Null hypothesis :

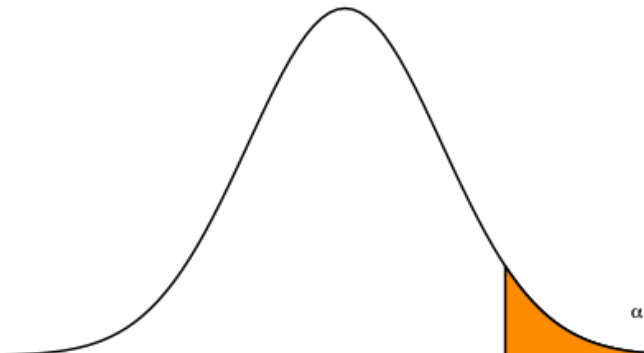
$\mu = 0$ - Alternative hypothesis 1 : $\mu > 0$ - Alternative hypothesis 2 : $\mu < 0$

Choosing One or Two Tails

Normal Distribution



Normal Distribution



Speaker notes

- One-tailed and two-tailed tests ask different questions, and are
- It is kind of like being ID when you're buy a beer – they're not asking, *Is this person different from 21?''* They're asking, *Is this person older than 21?''*

- “Do you feel lucky, punk? Well do ya?”

One-Tailed Test: Things to Consider

Before using a one-tailed test, ask yourself these questions:

1. Will the audience believe that I started with one tail before I saw the data?
 2. Will the audience share my opinion of which tail is interesting?
 3. Am I really 100 % committed to only this tail? - What if the effect turns out to be huge, but in the other direction?
- Would I be willing to call that a negative result?
 - Can I convince my audience I have this much commitment?

T-Test Assumptions

T-Test Assumptions, Part I

Assumptions of t-test

The textbook assumptions

- X is a metric variable.
- $\{X_1, X_2, \dots, X_n\}$ is a random sample.
- X has a normal distribution.

Variables are almost never normal.

Speaker notes

- And so, these assumptions are nearly never met in practice

T-Test Assumptions, Part II

But, in the large sample case, this is more plausible.

Large sample t-test assumptions

If: - X is a metric variable - $\{X_1, X_2, \dots, X_n\}$ is a random sample - n is large enough that the CLT implies a normal distribution of mean **Then:** The t-test is asymptotically valid

T-Test Assumptions, Part III

T-Test Assumptions, Part IV

The t-test is considered “reasonably robust,” even when $n < 30$, as long as deviations from normality are moderate.

However, watch out for strong skewness, especially when $n < 30$.

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- DAH: We’ve got to give students more to work with than this; I think that this might end up being unsatisfying for many folks.

T-Test Assumptions

More practical guidance:

- X is a metric variable.
- $\{X_1, X_2, \dots, X_n\}$ is a random sample.
- The distribution is not too non-normal, considering n .

When the t-test is not valid, consider using a non-parametric test instead.

Introduction to P-Values

Introducing P-Values

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- Let's note 2 things right away: just like in hypothesis testing.
- We're going to assume the null is true.
- We're measuring the probability of data - We may be interested in the probability that the null is correct - but statistics doesn't let us calculate that.

Z-Distribution

Speaker notes

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The P-Value for a Z-Test

Speaker notes

- $= 3 - 0/(20/\sqrt{100}) = 1.5$
- $p = 2 * (1 - \text{pnorm}(1.5)) = 0.13$

The P-Value and Decision Rules, Part I

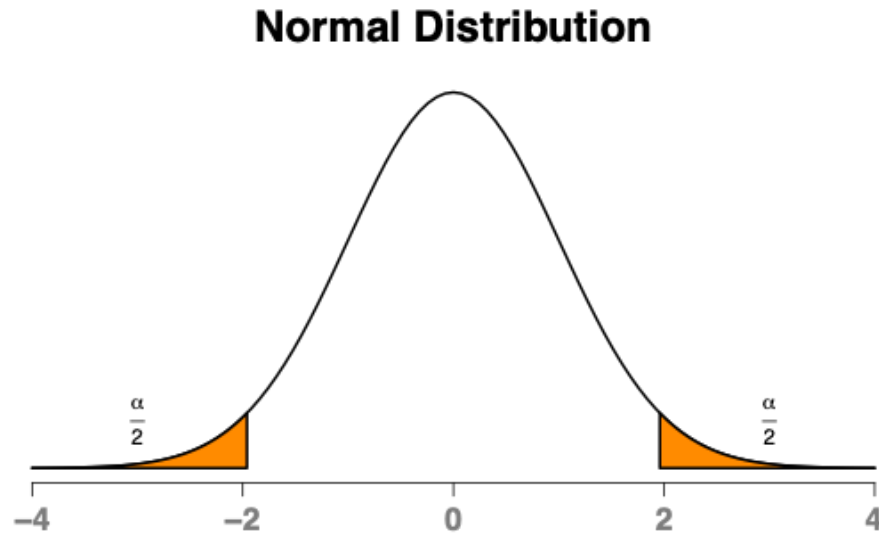
Neyman-Pearson hypothesis testing: rules to make a decision and usually be right ($\alpha = 0.05$)

- Strict frequentist with a dichotomous decision rule: treat $z = 2$ and $z = 10$ identically.
- But is there value in knowing *how contrary* the data is to the null?

The P-Value and Decision Rules, Part II

$|z| > \text{critical value} \Rightarrow \text{reject } H_0$

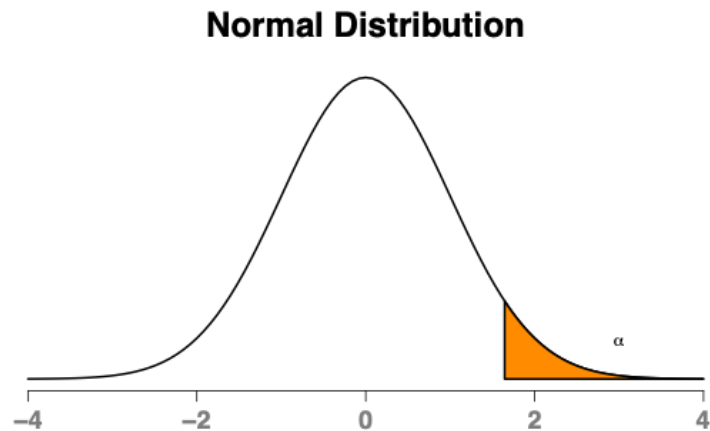
$|z| < \text{critical value} \Rightarrow \text{fail to reject } H_0$



The P-Value and Decision Rules, Part III

$|z| > \text{critical value} \Rightarrow \text{reject } H_0$

$|z| < \text{critical value} \Rightarrow \text{fail to reject } H_0$



An Equivalent Decision Procedure

Compute p-value.

- If $p < .05 \Rightarrow$ reject H_0
- If $p \geq .05 \Rightarrow$ do not reject H_0

But, can you justify making such a bright-line statement after reducing information so much?

1. Concept
2. Measurement
3. Statistic
4. Assumptions about distribution
5. **p-value**

t-Test and p-Values

P-Value Convention

Speaker notes

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Reporting Test Results

- A t-test for the effect of Vitamin W on blood pressure was highly significant ($t = 3.1$, $p = .008$).
- We found evidence that Vitamin W decreases blood pressure ($t = 2.3$, $p = .04$).
- The effect of Vitamin X on blood pressure was not statistically significant ($t = 1.2$, $p = .23$).

This is half the story; next, you'll need to describe practical significance.

Variable Importance and P-Values

Does a small p-value mean that a variable is “important”?

- Statistical significance
- Practical significance

A Warning

A very common mistake is to assume a p-value is the chance the null hypothesis is true.

Frequentist statistics cannot tell you the probability of a hypothesis!

A Warning (cont.)

Speaker notes

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Statistical Power

False Positive and False Negative Errors

- False Positive (I) errors are jumping without cause
- False Negative (II) errors are failing to jump when you should
 - Failing to detect a real effect
- Missed opportunity to create a product, publish a paper, or advance knowledge

Statistical Power, Part I

- False Negative Error Rate: $\beta = P(\text{not rejecting } H_0 | H_a)$
- Statistical power: $1 - \beta$
- Statistical power is the probability of supporting the alternate hypothesis, assuming it is true

Statistical Power, Part II

Speaker notes

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Statistical Power, Part II

Speaker notes

No notes on this slide.

Statistical Power, Part III

How to increase power

- Increase sample size.
- Choose a powerful test (if you can justify its assumptions).

Practical Significance

Practical significance

i Statistical significance

- How much does the data support the existence of an effect?

i Practical significance

- Is the size of this effect important?
- What is the magnitude of the effect?
- Should we care about this effect?

Speaker notes

- In many cases, statistical significance is little more than a statement of how large a sample was used to test a question.

Example

Speaker notes

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Practical Significance: Context

Primary goal

Provide context for your audience to reason about results.

- Who is your audience?
- What action might be taken based on these results?
- How does this result alter how you would run the business?
- What is the cost-benefit for implementing a change based on this result?
- How does this result “stack up” to other effects?

Speaker notes

- Understanding and communicating the business context is what separates good analysts from great data scientists.
- One might be very certain of a very small increase in a desirable metric; one might be less certain of a very large increase in a desirable metric.
- In many places, every relationship examined is significant. But, what do they
- We should directly tie this in to the machine learning conversation they’re going to have in 207 about feature importance.

Practical Significance: Model Explainability

- Some tasks require *explainable* models.
- Finance, healthcare, insurance, and other regulated industries stipulate specific model forms .
- Humans reason in linear hypotheses— higher-dimensional and conditional hypotheses are too much to keep in mind.

- This headline test feels out of place relative to the other content on this section.

Practical Significance: Effect Sizes

Effect sizes

- Single-number metrics that characterize the magnitude of an effect
- Population parameters that we estimate— *do not vary based on sample size*

Invalid effect size metrics- t-stat - p-value

Valid effect size metrics- Mean values - Difference in means between groups

- These provide general guidance about whether an effect is small, medium, or large.

Standard Effect Size Measures

Standardized effect sizes are designed to be flexible and apply in many scenarios:

- Cohen's d
- Correlation ρ
- Cramer's V

General metrics ignore the specific context around your research or business question.

Cohen's d

Sometimes, a mean (or difference in means) is hard to assess because the units are unfamiliar.

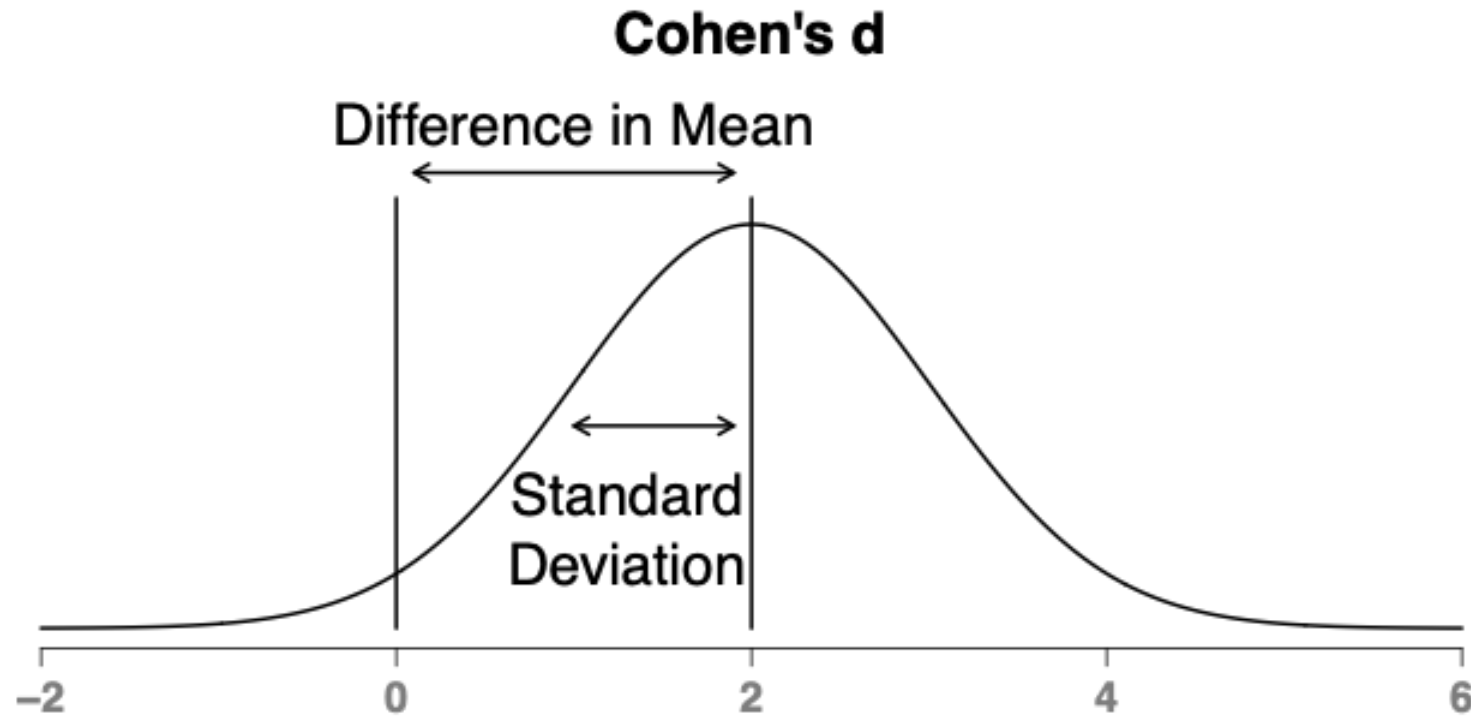
- **Example** : The effect of angled bristles on tooth decay is 5 millicaviparsecs per brushstroke %% TO THE REVIEWERS:
Yes, we know this isn't a word

Cohen's d

Compare effect size relative to the underlying natural variation in the outcome.

```
\[
  \text{Cohen's } d = \frac{\text{mean difference}}{\text{standard deviation}}
\]
```

Cohen's d (cont.)



 Rules of thumb (according to Cohen)

- Applicable across a huge number of contexts
- Ignores any important differences between context
- Saving dollars or saving lives are the same to Cohen's d

Takeaways

- After a statistical test, it's important to assess both statistical significance and practical significance.
- Standard effect size measures can help in a wide variety of situations.
- But don't get carried away and reach for them automatically.
- The main objective is to clearly explain how important the magnitude of the effect is.

Guidelines For Statistical Reporting

Guidelines for Statistical Reporting

- Communicating your results is a key part of statistical analysis
- In this class (and other classes in the program) we'll ask you to submit your analysis as a written report
- Next are some guidelines to keep in mind when writing a report

In this case, the guidelines are specific to exploratory analysis

Guideline One

Speaker notes

No notes on this slide.

Guideline Two

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Guideline Three

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Guideline Four

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Guideline Five

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Guideline Five (cont.)

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Guideline Six

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Guideline Six (cont.)

Speaker notes

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